

CONG WANG

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Ph.D. Candidate ◊ Microelectronic Thrust

The Hong Kong University of Science and Technology (Guangzhou)

RESEARCH INTERESTS

Electronic Design Automation (EDA), ML Accelerators, Domain Specific Architecture

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) *Sept. 2023 - Present*

Microelectronic Thrust

Supervisors: Prof. Shanshi Huang (Ph.D., Georgia Institute of Technology)

Prof. Wei Zhang (Ph.D., Princeton University; IEEE Fellow)

Ph.D. Candidate

Southern University of Science and Technology

Sept. 2021 - Jul. 2023

School of Microelectronic | *Supervisor: Prof. Quan Chen (Ph.D., The University of Hong Kong)*

M.Eng

Zhengzhou University

Sept. 2017 - Jul. 2021

School of Information Engineering

B.Eng

SELECTED PUBLICATIONS

- [1] **Cong Wang**, Liangcheng Zhao, Zexin Fu, Jing Zhang, Xunqin Lai, Jiayi Huang, and Shanshi Huang. “A Scalable Cluster-based Heterogeneous CIM Chiplet Architecture for Vision Transformers.” Under review.
- [2] Ruihao He, **Cong Wang**, Liangcheng Zhao, Xipeng Lin, Shaoxuan Li, and Hongwu Jiang. “Quantifying Scalability-Performance Trade-offs: A Design Space Exploration of Compute-in-Memory based Ising Machines.” *ACM Transactions on Design Automation of Electronic Systems*, 2026.
- [3] Shaoxuan Li, Zikun Wei, Ruihao He, **Cong Wang**, Ye Gu, Tianchu Dong, Yucheng Cai, and Hongwu Jiang. “TOKCIM: A 65nm All-Digital Compute-in-Memory Transformer Accelerator with Advanced Weight Sparsity Handling and SoftMax Acceleration.” *IEEE Custom Integrated Circuits Conference (CICC)*, 2026.
- [4] Yihang Zuo, Zexin Fu, **Cong Wang**, Yucuo Wu, Jiayi Huang, and Yuzhe Ma. “Harmony: A Hardware-Mapping Co-Exploration Framework for Hybrid CIM-based Vision Transformer Accelerator.” *International Symposium on Quality Electronic Design (ISQED)*, 2026. **Best Paper Award**.
- [5] Ruihao He, **Cong Wang**, Liangcheng Zhao, Zefeng Xu, and Hongwu Jiang. “Electronic-Photonic Hybrid CIM-based Ising Machine with Interaction-Aware Update Strategy for Combinational Optimization Problems.” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026.
- [6] Zhengxi Yan, **Cong Wang**, Xipeng Lin, Jing Zhang, and Hongwu Jiang. “CIM-SIFT: An Efficient Compute-in-Memory based Accelerator for Real-time SIFT Algorithm.” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026.
- [7] **Cong Wang**^{*}, Zexin Fu^{*}, Jiayi Huang, and Shanshi Huang. “Hemlet: A Heterogeneous Compute-in-Memory Chiplet Architecture for Vision Transformers with Group-Level Parallelism.” *arXiv preprint arXiv:2511.15397*, 2025. (^{*}Co-first authors)
- [8] **Cong Wang**, Zeming Chen, and Shanshi Huang. “MICSIm: A Modular Pre-Circuit Simulator for Mixed-Signal Compute-in-Memory Accelerators in CNNs and Transformers.” *Integration, the VLSI Journal*, 2025.
- [9] Ruihao He, **Cong Wang**, Xipeng Lin, Shaoxuan Li, and Hongwu Jiang. “RCIM: A Reconfigurable Compute-in-Memory Architecture for Artificial, Spiking and Hybrid Neural Networks.” *IEEE International Conference on Integrated Circuits, Technologies and Applications (ICTA)*, 2025.
- [10] Xipeng Lin, **Cong Wang**, Shanshi Huang, and Hongwu Jiang. “An Efficient Compute-in-Memory based Accelerator for Point-based Point Cloud Neural Networks.” *ACM/IEEE Design Automation Conference (DAC)*, 2025.
- [11] **Cong Wang**, Zeming Chen, and Shanshi Huang. “MICSIm: A Modular Simulator for Mixed-signal Compute-in-Memory based AI Accelerator.” *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.

- [12] **Cong Wang**, Dongen Yang, Jinming Lyu, Yong Dai, Cheng Zhuo, and Quan Chen. “On Model Order Reduction and Exponential Integrator for Transient Circuit Simulation.” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024.
- [13] **Cong Wang**, Dongen Yang, and Quan Chen. “EI-MOR: A Hybrid Exponential Integrator and Model Order Reduction Approach for Transient Power/Ground Network Analysis.” *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2022.

RESEARCH AND PROJECT EXPERIENCE

MICSim: Open-source CIM Accelerator Simulator *Sept. 2023 - Jul. 2024*

- First contributor; developed a modular simulator for mixed-signal CIM accelerators in CNNs and Transformers.
- Unified model-accuracy and hardware-performance evaluation; GitHub: MICSim_V1.0 has 39 stars.

Tiny TPU: MICS6000U Course Project *Fall 2024*

- Teaching assistant and lead contributor; implemented a compact TPU in SystemVerilog and authored course tutorials and cocotb testbenches.

Transient Circuit Simulation Acceleration *Jan. 2022 - Jul. 2023*

- Core contributor to an NSFC project; developed exponential-integrator methods to accelerate large-scale transient circuit simulation.

Model Order Reduction for Circuit Simulation *Jul. 2021 - Aug. 2022*

- Primary contributor to a Huawei HiSilicon collaboration; developed industrial-grade C++ programs for model order reduction methods.

HONORS AND AWARDS

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|---|---------------------|
| • HKUST(GZ) Research Postgraduate | <i>2023-present</i> |
| • SUSTech Outstanding Graduate | <i>2023</i> |
| • Best Paper Nomination, National Graduate Forum on Microelectronics | <i>2023</i> |
| • Third Place, Integrated Circuit EDA Elite Challenge | <i>2023</i> |
| • SUSTech Postgraduate Scholarship | <i>2021-2023</i> |
| • Zhengzhou University Third Prize Scholarship | <i>2021</i> |
| • National Encouragement Scholarship | <i>2019, 2020</i> |
| • Merit Student of Zhengzhou University | <i>2019, 2020</i> |
| • First Prize, Zhengzhou University National Mathematics Competition | <i>2020</i> |
| • Second Prize, Henan Province National Mathematics Competition | <i>2020</i> |
| • First Place, Zhengzhou University “Qiusi Cup” Mathematics Competition | <i>2020</i> |
| • First Prize, “Challenge Cup” National College Student Business Plan Competition | <i>2019</i> |
| • Excellent Student Cadre | <i>2019</i> |
| • Advanced Individual of Social Work | <i>2018</i> |
| • Excellent Student Volunteer | <i>2018</i> |

TEACHING

MICS6000U ML Accelerator, Teaching Assistant	<i>Fall 2024</i>
UCMP6010 Cross-disciplinary Research Methods, Teaching Assistant	<i>Spring 2025</i>

SKILLS

Programming: Python (NumPy, SciPy), C/C++, MATLAB.
Hardware & Verification: Verilog/SystemVerilog, cocotb; **AI Framework:** PyTorch.
Software & Tools: LaTeX, Markdown, MS Office; **Language:** Mandarin (Native), English (IELTS 7.0, CET-6).